
Listing 1: One model, composed and fitted. The caller states the model as a graph of named components and names a solver, or asks for `auto` to have the solver family derived from the graph; the parameter layout follows from the graph, and each kernel carries its own Jacobian, closed-form for all but eleven. The same construction scales to several components, tied parameters, stacked datasets and N-dimensional fits without a change of interface.

```
from spectrafit_core import MeasurementData, compose, fit, gaussian

# x, y are the measured abscissa and ordinate.
graph = compose([gaussian("peak1", amplitude=1.0, center=0.0, sigma=1.0)]).build()
result = fit(graph, MeasurementData(x=list(x), y=list(y)))

result.parameters      # keyed "peak1.<param>"; each carries .value and .stderr
result.r_squared       # goodness of fit, as  $R^2$  against the data
```
